SAKSHAM SINGH SACHAN

B.Tech Computer Science and Engineering Student

Gmail | GitHub | LinkedIn | Twitter | Leetcode | Codeforces | Website

Education

B.Tech in Computer Science and Engineering
Present
CSJMU UIET (formerly Kanpur University)

Achievements

- Lichess rating of approximately **2300–2400**
- Chess.com rating of approximately **1930–2000**

Technical Skills

Programming Languages: JavaScript, TypeScript, Rust, Python, C/C++, Java, Bash, Dart/Flutter, HTML/CSS
Frameworks & Tools: React, Svelte, Astro, Tailwind CSS, Node.js, Express, SQL/NoSQL (MongoDB), Docker, Kubernetes, Git, NGINX, Caddy
Cybersecurity: Offensive Security, Application Security (Web/API), Vulnerability Research, Threat Intelligence, OSINT, Attack Surface Management (ASM), Threat Hunting, Secure Code Review, Authentication & Authorization Testing, Business Logic Testing, API Security, IDOR & Access Control Analysis, Reconnaissance Automation, Detection Engineering, MITRE ATT&CK, Burp Suite, Nmap, ffuf, Nuclei, Metasploit, Wireshark, Ghidra
Operating Systems: Linux (Arch, Debian, Ubuntu, Kali), macOS, Windows, Android, Embedded Linux; Proficient with Linux internals, system administration, networking, shell scripting, virtualization (Docker/UTM), and CLI-based development workflows.
DevOps & Infra: GitHub Actions, Fly.io, DigitalOcean, GCP, CI/CD, Railway, Heroku
Database & ORMs: MongoDB, SQLite, Firebase, MySQL, Postgres, Drizzle, Prisma
Areas of Interest: Web Application Security, Ethical Hacking, Vulnerability Research, Security Automation

Certifications

- **Arch1001: x86-64 Assembly** – OpenSecurityTraining2 *Issued Jul 2026*
- **Ethical Hacker** – Cisco *Issued Jun 2026*
- **Introduction to Cybersecurity** – Cisco *Issued Jun 2026*

Security Research & Vulnerability Disclosure

Independent Web Security Research – Multiple University Web Properties

- Conducted independent security research across web properties of **Delhi Technological University (DTU), Lovely Professional University (LPU), and CSJMU**, identifying security weaknesses through systematic analysis of application behaviour, HTTP security controls, and exposed attack surfaces.
- Performed structured assessment of web security configurations, analysing security-relevant HTTP response headers, framing policies, and application-level controls to identify weaknesses with potential security implications.
- Independently validated identified security issues, evaluated their potential impact and attack conditions, and documented reproducible technical findings suitable for security triage and remediation.
- Prepared vulnerability reports containing technical observations, affected assets, security impact, validation evidence, and remediation considerations, and submitted the findings through the **CERT-In vulnerability reporting channel** for responsible disclosure.

Vulnerability Disclosure – IIT Kanpur Website

- Identified and responsibly disclosed a security vulnerability affecting an IIT Kanpur web property through independent security research.
- Conducted technical analysis of the affected web property, validated the security issue, assessed its potential impact, and documented the finding for responsible disclosure.
- The vulnerability report was acknowledged by an IIT Kanpur professor, providing external recognition of the reported security finding.

Physics & Space Science

Research Interests: Physics, Astrophysics, Space Science, Earth Observation, Atmospheric Science, Computational Physics

Selected for ISRO Training Programmes

- **AI/ML for Geodata Analysis** *ISRO — Aug 03–14, 2026*
- **Earth Observations and Numerical Model Applications for Tropical Cyclone Monitoring and Forecasting** *ISRO — Jul 30, 2026*
- **Aerosols: Measurement, Retrieval and Impacts** *ISRO — Jul 27–28, 2026*

Interests: Exploring the intersection of physics, computational methods, artificial intelligence, and space science, with a particular interest in astrophysics, atmospheric phenomena, and Earth observation.

Projects

SSOS – Custom Operating System *Systems Programming — C/C++*

- Developing a custom operating system from scratch to gain hands-on experience in low-level systems programming, computer architecture, and kernel development.
- Working on core operating-system concepts including kernel development, memory management, process management, interrupts, system calls, and hardware-level interaction.
- Exploring the complete software stack from system boot and low-level hardware interaction to higher-level operating-system abstractions.

Website Security Scanner *Python — Cybersecurity*

- Developed a Python-based security scanning tool to automate website reconnaissance and identify potential security weaknesses.
- Implemented automated security checks and reconnaissance functionality to streamline the initial assessment of web applications.
- Designed the project to strengthen practical understanding of web application security, HTTP communication, vulnerability assessment, and security automation.

Instagram OSINT Tool *Python — OSINT — Open Source*

- Developed a Python-based OSINT tool focused on gathering and analysing publicly available information associated with Instagram profiles.
- Automated repetitive information-gathering workflows to improve the efficiency of open-source intelligence investigations.
- Designed the project to explore practical OSINT methodologies, data collection, and Python-based automation, and published it as an open-source GitHub project.

AI-Based Resume Shortlisting System *AI — Python — Web Application*

- Developed a prototype AI-powered resume shortlisting system designed to assist recruiters in analysing candidate resumes and identifying relevant applicants.
- Implemented an automated resume-analysis workflow to evaluate candidate profiles against recruitment requirements and support the shortlisting process.
- Built a functional prototype with an interactive user interface and integrated AI functionality, with the potential for further development into a complete recruitment platform.

Rocket Landing Simulator *Python — Simulation — Physics*

- Developed a rocket landing simulation inspired by real-world autonomous rocket landing systems, focusing on the computational modelling of a controlled landing trajectory.
- Implemented simulation logic to model rocket motion and analyse the effect of control parameters on the landing process.
- Explored concepts related to physics-based simulation, trajectory control, and autonomous landing systems through an interactive computational environment.

Open Source

- Maintaining a GitHub portfolio focused on cybersecurity, Python development, OSINT and security automation.
- Building practical cybersecurity projects and security-focused utilities.
- Interested in contributing to open-source cybersecurity projects.

Activities

- Creating beginner-friendly cybersecurity content covering CTF walkthroughs and cybersecurity concepts.
- Developing faceless short-form educational content for students interested in cybersecurity.
- Exploring cybersecurity, OSINT, open-source development and security research.

Relevant Coursework

Data Structures and Algorithms	Computer Networks	Operating Systems
Database Management Systems	Computer Architecture	Object-Oriented Programming
Web Technologies	Software Engineering	Cybersecurity Fundamentals